Beating Hallucinations via Uncertainty-Conditioned Attention Gating

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Abstract

Large Language Models (LLMs) frequently hallucinate in abstractive summarization and open-domain question answering, undermining reliability in factuality-critical applications. Existing methods rely on output-based evaluation or post-hoc verification, offering little insight into the internal mechanisms that produce hallucinations or which components causally drive them.

We propose an inference-time framework that detects and mitigates hallucinations using model-internal signals alone, requiring no external retrieval or multiple decoding passes. Our approach combines a Semantic Entropy Probe (SEP) trained on hidden states to estimate hallucination risk with the Lookback Ratio (LR), which measures each attention head's reliance on input context. When the SEP signals high uncertainty, a head-adaptive gate selectively suppresses poorly grounded attention heads. We evaluate on five benchmarks (XSum, NaturalQuestions, TriviaQA, SQuAD, HaluEval) using Llama-2-7B, reporting AUROC, token-level F1, and accuracy, and demonstrate that combining uncertainty and grounding signals yields effective single-pass hallucination detection and mitigation.

1 Introduction

Large Language Models (LLMs) have rapidly advanced natural language processing, demonstrating strong performance on summarization, question answering, and reasoning. Despite this, LLMs frequently hallucinate, generating outputs that are factually incorrect or unsupported by the input context (Yu et al., 2024; Chuang et al., 2024a) - limiting their reliability in high-stakes domains such as healthcare, journalism, and knowledge retrieval (Wang et al., 2025; Kossen et al., 2024).

We focus on hallucination in open-domain question answering and abstractive summarization, where outputs must be grounded in the input context or external knowledge. LLMs frequently bypass this grounding by relying on spurious correlations or parametric memory (Yu et al., 2024; Chuang et al., 2024a). Detecting such failures is non-trivial: hallucinated outputs can appear fluent and plausible, and standard metrics like ROUGE do not capture factual consistency (Yu et al., 2024). Crucially, hallucinations can arise at intermediate generation steps before manifesting in the final output, making post-hoc detection insufficient and motivating inference-time intervention (Wang et al., 2025; Kossen et al., 2024).

Existing hallucination detection methods largely operate at the output level using reference-based metrics or external verifiers, providing no insight into the internal mechanisms driving errors and precluding real-time mitigation. Recent work has explored internal signals such as semantic entropy over multiple generations (Kossen et al., 2024), and attention-based context reliance (Chuang et al., 2024a), as well as mechanistic interpretability analyses of specific attention heads and MLP layers (Yu et al., 2024). However, these approaches are studied in isolation and remain largely correlational. Mitigation methods based on attention modulation (Wang et al., 2025; Tong et al., 2025) often apply head-level interventions unconditionally, incurring unnecessary overhead on confident samples or requiring architectural changes that limit deployment in latency-sensitive settings.

We propose an inference-time framework that combines a per-sample uncertainty trigger with per-head grounding suppression. A lightweight Semantic Entropy Probe (SEP) trained on hidden states estimates hallucination risk (Kossen et al., 2024). When uncertainty is high, a Lookback Ratio (LR)-based gate (Chuang et al., 2024a) selectively suppresses attention heads with low context reliance, leaving confident generation untouched. We show that combining SEP with LR-based gating improves hallucination detection over either signal alone, and that this conditional design outperforms always-on head modulation (Tong et al., 2025) while preserving output quality on confident samples. We further conduct structured ablation and blindness tests (Yu et al., 2024) to provide causal evidence for which specific attention heads drive hallucinated outputs, going beyond the correlational analyses of prior work.

We evaluate on five benchmarks spanning summarization and question answering: XSum (Narayan et al., 2018), Natural Questions (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), SQuAD (Rajpurkar et al., 2016), and HaluEval (Li et al., 2023a), using Llama-2-7B. Following Kossen et al. (2024), we train SEP probes across all hidden layers and identify those providing the strongest hallucination signal, reporting AUROC, token-level F1, and accuracy throughout.

Contributions.

- **Unified Internal-Signal Framework:** We combine Semantic Entropy Probes (SEP) (Kossen et al., 2024) and Lookback Ratio (LR) (Chuang et al., 2024a) into a single-pass inference-time framework. We show that these signals capture complementary aspects of hallucination risk and that their combination outperforms either signal alone.

- **Uncertainty-Conditioned Inference-Time Intervention:** We introduce a conditional attention intervention that activates only under high semantic entropy (Kossen et al., 2024), selectively suppressing risky attention heads rather than modulating them always-on as in HAVE (Tong et al., 2025). This reduces hallucinations during uncertain states while leaving confident generation intact.

- **Causal Analysis of Hallucination Mechanisms:** Drawing on mechanistic interpretability (Yu et al., 2024), we conduct structured ablation and blindness tests to establish that identified attention heads *causally* contribute to hallucinations and not merely correlate with them, by showing that suppressing these heads directly alters hallucinated outputs.

- **Empirical Characterization of Hallucination Heads:** We show that hallucination-associated heads consistently exhibit lower Lookback Ratios (Chuang et al., 2024a) and higher SEP-captured entropy (Kossen et al., 2024), yielding an interpretable internal signature of how hallucinations arise during generation.

2 Related Work

2.1 Hallucination Detection and Uncertainty Estimation

Hallucination detection in LLMs has largely relied on output-based evaluation and uncertainty estimation. Metrics such as ROUGE fail to capture factual consistency in abstractive tasks. A foundational line of work estimates uncertainty via semantic entropy, which clusters generations by meaning and measures variability in semantic space (Farquhar et al., 2024). A related approach, self-consistency, samples multiple reasoning paths and returns the most frequent answer, improving factual accuracy without additional training (Wang et al., 2023). To reduce the multi-generation cost of these methods, Semantic Entropy Probes (SEP) predict semantic entropy directly from hidden states, demonstrating that hallucination signals are internally encoded within the model (Kossen et al., 2024). Despite their effectiveness, these methods remain largely correlational and often require post-hoc analysis, limiting applicability in real-time settings.

2.2 Attention-Based Grounding and Interpretability

Attention-based methods analyze how LLMs utilize context during generation. The Lookback Ratio measures attention to source tokens versus generated tokens, showing that reduced context reliance correlates with hallucination (Chuang et al., 2024a). Extensions include attention editing methods that modify attention maps during inference (Wang et al., 2025) and head-level gating approaches that dynamically reweight attention contributions (Tong et al., 2025) to improve grounding.

A complementary direction intervenes directly on internal activations. Inference-Time Intervention (ITI) identifies attention heads encoding truthful directions and shifts activations along those directions at inference, improving truthfulness without retraining (Li et al., 2023b). DoLa improves factuality by contrasting next-token distributions from later versus earlier layers, exploiting the observation that factual knowledge is localized in higher transformer layers (Chuang et al., 2024b).

Mechanistic interpretability studies further identify internal components responsible for model behavior. Techniques such as logit lens and causal mediation analysis link specific attention heads and MLP layers to output predictions (Yu et al., 2024).

2.3 Evaluation Benchmarks for Faithfulness and QA

Hallucination evaluation has been studied across multiple benchmarks in both summarization and question answering. In summarization, highly abstractive datasets are commonly used because they encourage models to generate novel content, increasing hallucination risk (Chuang et al., 2024a; Wang et al., 2025). Faithfulness evaluation remains challenging, as overlap-based metrics such as ROUGE do not fully capture factual accuracy (Yu et al., 2024).

In question answering, standard extractive and open-domain benchmarks evaluate correctness using exact match and F1, but these metrics do not explicitly capture hallucination when models generate plausible but unsupported answers (Kossen et al., 2024; Yu et al., 2024). More recent work has proposed uncertainty-based detection (Farquhar et al., 2024; Kossen et al., 2024), attention-based grounding methods (Chuang et al., 2024a), and decoding-time interventions (Li et al., 2023b; Chuang et al., 2024b) as complementary evaluation and mitigation strategies.

3 Methodology

3.1 Overview

We leverage single-pass internal signals for detection and mitigation of hallucinations at inference time, moving beyond post-hoc or multi-sample approaches, by integrating grounding signals (Chuang et al., 2024a) with uncertainty estimates derived from Semantic Entropy Probes (Kossen et al., 2024).

Additionally, most hallucination studies focus on a single task, limiting understanding of how hallucinations generalize across domains. We perform a unified evaluation across summarization and question answering benchmarks and focus on inference-time mitigation using our proposed framework.

We combine hallucination detection with targeted attention intervention. First, we label data using semantic entropy computed from clustered stochastic generations (Kossen et al., 2024), and train a lightweight SEP on hidden states to predict hallucination risk. Next, we extract LR features from attention to measure how strongly each head relies on the input context (Chuang et al., 2024a). Finally, we apply a head-adaptive gating mechanism that suppresses low-grounded attention heads during inference (Tong et al., 2025), triggered only when the SEP indicates high hallucination risk.

3.2 Semantic Entropy Probe

In order to prepare the dataset for training our Semantic Entropy Probe (SEP), we sample 2000 examples from datasets such as XSum or SQuAD. Next, we generate 10 stochastic summaries per example using the Meta Llama2-7B model. Then, we use DeBERTa as an NLI model to cluster the generated answers by semantic equivalence following prior semantic entropy estimation procedures (Kossen et al., 2024). Samples showing bidirectional entailment are assigned to the same cluster. Further, we calculate the semantic entropy over these clusters. Samples with high agreement ($>$ 0.7) are labeled Confident (0), while those with low agreement ($<$ 0.3) are labeled Hallucinated (1). Ambiguous samples are discarded.

Then, we perform a prompt-only forward pass on each filtered sample and extract the hidden state of the last token from each transformer layer. We then train an L2-regularized logistic regression classifier on the normalized hidden state features, in order to predict hallucinations. The obtained classifier is evaluated on the rest of the dataset, and obtain the best contiguous series of layers (>= 5) which give the best AUROC score in detecting hallucinations.

3.3 Lookback Ratio Feature Extraction

We extract the lookback ratio (LR) to quantify how strongly each attention head grounds its predictions in the input context (Chuang et al., 2024a). For each sample, we run the model with attention outputs enabled, collecting attention weights at every generation step. At each layer l and attention head h, we compute the proportion of attention mass assigned to prompt tokens relative to the total attention mass over all tokens (prompt + generated), yielding a per-step grounding score. These scores are averaged across all generated tokens to obtain a stable estimate $\mathrm{LR}_{l,h} \in [0, 1]$, where higher values indicate stronger reliance on the input context. The resulting per-head scores are flattened into a fixed-length feature vector of size L x H for each sample. For each SEP-triggered sample, the LR is used to drive inference-time interventions that selectively

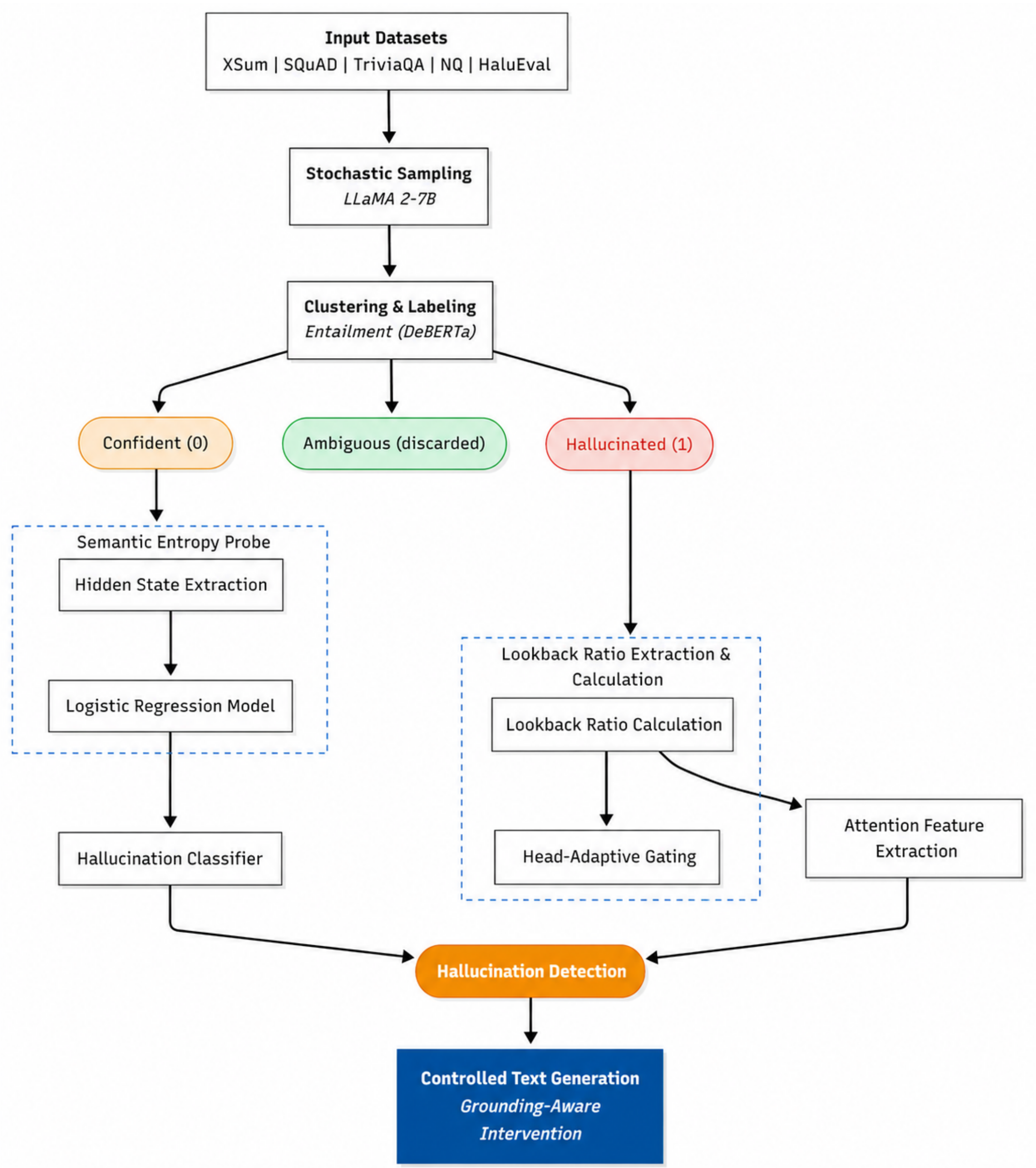

Figure 1: **Overview of our hallucination mitigation pipeline.** Proposed three-stage pipeline: (1) SEP for hallucination detection, (2) LR for estimating attention grounding, and (3) Head-adaptive gating to suppress ungrounded attention heads during generation.

suppress low-lookback (i.e., less grounded) attention heads.

3.4 Head-Adaptive Gating Using Lookback Ratio

We introduce a head-adaptive gating mechanism that selectively modulates attention heads during generation based on their grounding behavior, inspired by prior head-level modulation strategies (Tong et al., 2025). Specifically, at each decoding step, we compute the lookback ratio $\mathrm{LR}_{l,h}$ for every attention head. This produces a per-head grounding score in [0,1], where higher values indicate stronger reliance on the input context. We then transform this score into a multiplicative gate applied directly to the head's contribution to the attention output. We introduce two variants of this multiplicative gate - a soft variant, and a hard variant. In the soft variant, the gate is defined as:

$$g_{l,h} = \sigma\big((\mathrm{LR}_{l,h} - \tau) \cdot \alpha\big)$$

where τ is a cutoff threshold and α controls sharpness; this centers the gating function around the cutoff, suppressing low-lookback (ungrounded) heads while preserving high-lookback (grounded) ones. In the hard variant, heads are either fully retained or suppressed based on whether $\mathrm{LR}_{l,h} \geq \tau$. The gating is applied immediately after the attention softmax and before the output projection, ensuring that each head's contribution is modulated independently without altering the attention distribution itself. To minimize unnecessary intervention, this mechanism is activated only when a lightweight semantic entropy probe predicts high hallucination risk, enabling targeted, inference-time control over attention flow.

4 Experimental Setup

4.1 Datasets

We evaluate our approach on five widely used NLP benchmarks spanning summarization, question answering and hallucination detection. XSum is an abstractive summarization dataset consisting of BBC news articles paired with single-sentence summaries, designed to test extreme compression and factual consistency. Natural Questions (NQ) is a question answering dataset derived from real user queries issued to a search engine, with answers grounded in Wikipedia passages, emphasizing open-domain QA. TriviaQA contains trivia-style questions with evidence documents from the web, requiring multi sentence reasoning and answer extraction. SQuAD is a reading comprehension dataset based on Wikipedia articles, where models must extract answer spans from given contexts, focusing on precise span-level understanding. HaluEval is a hallucination evaluation benchmark, containing annotated examples of factual and non-factual generations across multiple tasks. We use its QA subset to measure whether head-adaptive gating reduces hallucinated responses while maintaining grounded answer accuracy. These datasets collectively enable evaluation of hallucination behavior across both generative summarization and extractive/open-domain question answering settings.

Dataset	Source	Domain	Size
XSum	BBC	News	∼227K
NQ	Google + Wiki	QA	∼307K
TriviaQA	Trivia + Wiki	QA	∼95K
SQuAD	Wiki	RC	∼100K
HaluEval	OpenAI	Hallucination Eval	∼35K

Table 1: Datasets used in our experiments.

4.2 Models

We conduct our experiments using two open-weight large language models from the LLaMA family. Specifically, we use LLaMA 2-7B and LLaMA 3-8B-Instruct for both the abstract summarization and question-answering tasks. The models are used in a few shot inference setting, where they are provided 5 question-answer (or 3 summarization) examples along with the actual question/data to summarize.

4.3 Evaluation Metrics

We evaluate our method using a combination of generation quality and detection metrics. ROUGE is used to approximate semantic consistency in the context of semantic entropy. To select the most informative hidden layer for the Semantic Entropy Probe (SEP), we use AUROC, which measures the model's ability to discriminate between hallucinated and factual outputs. For downstream evaluation, we report token-level F1 score to capture the balance between precision and recall at the token level, along with accuracy to measure exact token prediction correctness.

4.4 Implementation Details

We use a few-shot prompting strategy with all evaluated models, with 5 examples given for each QA task generation, and 3 examples given for every summarization task generation. For sampling question-answer pairs from the selected datasets for training the SEP, we use a random seed of 20. We set max_new_tokens = 50 for generating responses for evaluation on QA tasks, and max_new_tokens = 100 on summarization tasks. When models are used in a high temperature setting, the temperature is set to 1.0. Otherwise, if they are used in a low-temperature setting, it is set to 0.1.

5 Results and Discussion

5.1 Cross-Dataset Performance

Objective. This experiment evaluates whether the proposed Semantic Entropy Probe (SEP) learns

Model	Architecture	Params	Mode
LLaMA 2-7B	Decoder-only Transformer	7B	Few-shot
LLaMA 3-8B-Instruct	Decoder-only Transformer	8B	Few-shot

Table 2: Models used in our experiments.

Metric	Measures
ROUGE	N-gram overlap for semantic entropy estimation
AUROC	Discrimination ability for SEP layer selection
Token-level F1	Precision/recall balance at token level
Accuracy	Exact token prediction correctness

Table 3: Evaluation metrics used in our study.

Train \Eval	SQuAD	TriviaQA	NQ
SQuAD	**0.7622**	0.7488	0.7428
TriviaQA	0.6830	**0.7705**	0.7162
NQ	0.7155	0.7302	**0.7929**

Table 4: Cross-dataset AUROC for hallucination detection using SEP. Diagonal entries denote in-distribution performance, while off-diagonal entries correspond to out-of-distribution transfer.

Train $\setminus$ Eval	XSum	HaluEval-QA
XSum	**0.5491**	0.5293
HaluEval-QA	0.4750	**0.7406**

Table 5: Cross-dataset AUROC for hallucination detection using SEP on datasets for differing tasks - hallucination detection/QA vs summarization.

generalizable hallucination signals that transfer across datasets. Specifically, we assess whether SEP trained on one dataset maintains strong performance when applied to out-of-distribution datasets, and quantify the extent of performance degradation under cross-domain transfer.

Findings. We evaluate the generalization of the Semantic Entropy Probe (SEP) using cross-dataset AUROC matrices. When trained and evaluated on the same dataset (in-distribution), SEP achieves a mean AUROC of 0.7752. Under cross-dataset transfer (out-of-distribution), performance degrades moderately to an average AUROC of 0.7227. While some transfer pairs (e.g., SQuAD to TriviaQA) retain strong performance, others exhibit larger drops, indicating partial domain sensitivity. Across these datasets, our method demonstrates strong generalization (OOD/ID AUROC ratio of 0.93). We further evaluate the generalization of the Semantic Entropy Probe (SEP) under cross-task transfer by training on one task domain and evaluating on another (summarization $\leftrightarrow$ question answering).

Table 5 reports the cross-dataset AUROC matrix between XSum and HaluEval-QA. In the in-distribution setting, SEP achieves a mean AUROC of 0.6448 across the diagonal entries. However, under cross-task out-of-distribution transfer, performance drops substantially to an average AUROC of 0.5022, corresponding to an OOD/ID ratio of 0.7788. Notably, training on XSum transfers weakly to HaluEval-QA (0.5293), while HaluEval-QA fails to generalize to XSum (0.4750), approaching random performance.

These results indicate that although SEP captures useful hallucination signals within task domains, its learned decision boundary is only weakly transferable across fundamentally different generation tasks, suggesting that semantic entropy thresholds and probe calibration remain task-dependent.

5.2 Ablation Study

Objective. The ablation study evaluates the causal role of grounding-aware attention heads and the effectiveness of head-adaptive gating. Specifically, it investigates (1) whether selectively suppressing low lookback-ratio heads improves robustness compared to indiscriminate removal, and (2) whether high lookback-ratio heads are causally responsible for accurate, grounded predictions.

Findings. Results show that soft, sigmoid-based gating consistently outperforms or matches hard knockout of all heads, indicating that selective suppression preserves useful information while mitigating ungrounded contributions. Knockout experiments demonstrate that indiscriminate removal of attention heads leads to larger performance degradation, supporting the need for fine-grained control. Blindness tests further reveal that removing high lookback-ratio heads significantly reduces accuracy on multiple datasets, confirming their causal role in grounding model predictions. Overall, the findings validate the effectiveness of head-adaptive gating

Dataset	Orig (Trig)	Gate	Knockout	Orig (Pass)	Blind	Observation
TriviaQA	0.25	0.25	0.19	0.83	0.83	Soft > Hard; Blindness inconclusive
SQuAD	0.09	0.07	0.08	0.44	0.37	Hard ≈ Soft; Grounding causal
NQ	0.11	0.11	0.07	0.53	0.50	Soft > Hard; Grounding causal
HaluEval-QA	0.60	0.58	0.55	0.88	0.87	Soft > Hard; Grounding causal
XSum	0.50	0.49	0.42	0.55	0.57	Soft > Hard; Blindness inconclusive

Table 6: Causal validation results of head-adaptive gating. "Orig (Trig)" denotes accuracy on SEP-triggered samples before gating, "Gate" corresponds to sigmoid gating, and "Knockout" denotes hard removal of all heads. "Orig (Pass)" and "Blind" represent accuracy before and after zeroing high-LR heads on SEP-passthrough samples.

and highlight the importance of grounding-aware attention in hallucination mitigation.

5.3 Discussion

Our results demonstrate that hallucination in large language models can be effectively detected and mitigated using internal signals without requiring external verification. SEPs provide a lightweight and generalizable mechanism for identifying hallucination risk from hidden states, while LR features reveal interpretable grounding patterns at the attention-head level. Together, these components enable targeted intervention via head-adaptive gating, which selectively suppresses ungrounded attention heads. Across datasets, we observe that this combination maintains strong performance while improving robustness, highlighting the importance of internal model dynamics in controlling hallucinations.

These findings suggest a shift from post-hoc verification toward inference-time control of model behavior. By leveraging signals already present within the model, our approach avoids the overhead of additional models or external knowledge sources, making it efficient and broadly applicable. The use of attention-based grounding signals further opens avenues for interpretable and controllable generation, where specific components of the model can be modulated based on predicted risk. This points to a promising direction for building more reliable language models through internal monitoring and adaptive computation.

Limitations

- **Dataset limitations:** Our experiments are limited to English-language datasets (XSum, SQuAD, TriviaQA, NQ, HaluEval), which primarily consist of news and Wikipedia-based content. These datasets do not capture low-resource languages, diverse dialects, or domain-specific knowledge, and may exhibit annotation noise or bias. Additionally, the sampled subset used for semantic entropy labeling is relatively small, which may limit coverage of rare or complex hallucination cases.
- **Methodological limitations:** Our approach relies on semantic entropy as a proxy for hallucination, which assumes that agreement across sampled generations correlates with factual correctness. This assumption may not always hold, particularly for ambiguous or underspecified queries. Furthermore, lookback ratio focuses only on attention patterns and does not capture all sources of hallucination, such as parametric knowledge errors or reasoning failures.
- **Model limitations:** We evaluate our method on open-weight models up to 8B parameters (LLaMA 2-7B and LLaMA 3-8B-Instruct). The behavior of the proposed approach on larger or proprietary models remains unexplored due to insufficient compute resources and access limitations. In addition, our method requires access to internal activations and attention weights, which may not be available in closed-source or API-only systems.
- **Computational limitations:** Extracting semantic entropy labels requires multiple stochastic generations per sample, and lookback ratio computation requires attention outputs during decoding, both of which increase computational cost. As a result, experiments are conducted on limited subsets and model sizes, and full-scale evaluations on larger datasets or models are constrained by hardware availability. Additionally, experiments could not be run using other summarization benchmarks like CNN-DailyMail and others, due to the length of the examples in the data exceeding the capability of our hardware.
- **Generalizability:** While our method shows promising cross-dataset performance within

question answering and summarization tasks, its effectiveness in other settings such as dialogue, code generation, or highly specialized domains remains untested. Additionally, generalization to multilingual or low-resource scenarios is unclear, and may require further adaptation of both the entropy-based labeling and attention-based features.

- **Generalization quality:** Fluency and coherence of gated outputs are not fully evaluated, leaving open the question of whether accuracy gains come at a cost to linguistic quality.

6 Conclusion

We present a lightweight, inference-time framework for hallucination detection and mitigation that combines a Semantic Entropy Probe (SEP) with grounding-aware attention control. SEP predicts hallucination risk from hidden states, while lookback ratio (LR) features quantify how strongly attention heads rely on the input context. We use these signals to apply head-adaptive gating, selectively suppressing ungrounded attention heads during generation. Across five datasets, our method demonstrates strong generalization for tasks in the same domain (OOD/ID AUROC ratio of 0.93 on QA benchmarks), and weak generalization in a cross-task setting (OOD/ID AUROC ratio of 0.78 between summarization and QA). We also demonstrate causal effectiveness, showing that grounding-aware attention modulation can improve reliability without degrading performance.

Our results highlight the value of internal model signals for efficient and interpretable control of generation, offering an alternative to post-hoc or external verification approaches.

The cross-task failure of SEP suggests that task-specific probe training is currently necessary. Future work should explore whether a single task-agnostic probe can be trained using multi-task semantic entropy labels.

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A Full Prompt Templates

We provide the exact prompt templates used for generation to ensure reproducibility.

Natural Questions (NQ).

```
Answer the following question as briefly as
    possible.
Question: when was the first gas powered
    airplane invented?
Answer: 1903

Question: what was the original name of
    jacksonville florida?
Answer: Fort Caroline

Question: damage to this part of the brain
    can result in reduced ability to
    comprehend and/or produce humor?
Answer: right hemisphere

Question: what does the green on the torres
    strait islander flag represent?
Answer: the land

Question: who are the democrats running for
    governor of texas?
Answer: Lupe Valdez
```

SQuAD.

```
Answer the following question as briefly as
    possible.
Question: Who is one of the largest
    landowners in the UK?
Answer: The Ministry of Defence

Question: What adverse effect was
    thalidomide associated with?
Answer: congenital abnormalities

Question: When was the Lorelei Fountain
    written about?
Answer: 1838

Question: Who starred in the film Three
    Smart Girls?
Answer: Deanna Durbin

Question: What was the name of Madonna's
    eleventh album?
Answer: Hard Candy
```

TriviaQA.

```
Answer the following question as briefly as
    possible.
Question: What is the first name of James
    Bond villain Dr No?
Answer: Julius

Question: Who was crowned MotoGP World
    Champion after he finished third in the
    Malaysian Grand Prix on 10 October 2010?
Answer: Jorge Lorenzo

Question: What German battleship was sunk in
    November 1944 by Barnes-Wallis' Tallboy
    bombs, dropped by Lancaster bombers?
Answer: Tirpitz

Question: Which TV personality has written
    the best-selling autobiography 'My
    Animals and Other Family'?
Answer: Clare Balding

Question: Which of the seven wonders of the
    ancient world was thought to have
    straddled the harbour of a Greek island?
Answer: Colossus of Rhodes
```

XSum.

```
Summarize the following article in one
    sentence.

Article: [ARTICLE TEXT]
Summary:

Article: [ARTICLE TEXT]
Summary:

Article: [ARTICLE TEXT]
Summary:
```

HaluEval.

```
Context: Kevin Byrne is the Assembly member
    for the 94th District of the New York
    State Assembly. The district includes
    portions of Putnam and Westchester
    counties in the Hudson Valley. Putnam
    County formed in 1812 from Dutchess
    County and is named for Israel Putnam, a
    hero in the French and Indian War and a
    general in the American Revolutionary
    War.
Question: What county named for a hero in
    the French and Indian War and a general
    in the American Revolutionary War is
    represented by Assemblyman Kevin Byrne?
Answer: Putnam County

Context: EuropaCorp is a French motion
    picture company headquartered in
    Saint-Denis, a northern suburb of Paris,
    and one of a few full-service
    independent studios that both produces
```

```
     and distributes feature films, as well
     as one of the major companies in Europe.
     The film industry or motion picture
     industry comprises the technological and
     commercial institutions of filmmaking,
     i.e., film production companies, film
     studios, cinematography, animation, film
     production, screenwriting,
     pre-production, post-production, film
     festivals, distribution, and actors,
     film directors, and other film crew
     personnel.
Question: What kind of company headquartered
     in Saint-Denis comprises the
     technological and commercial
     institutions of filmmaking?
Answer: motion picture company

Context: Around the start of the season, the
     Timberwolves would be the first team in
     NBA history to hold four players that
     were around 20 or younger: Andrew
     Wiggins, Zach LaVine, Karl-Anthony
     Towns, and Tyus Jones. He played college
     basketball for the University of
     Kentucky.
Question: Which Minnesota Timberwolves
     player was around 20 or younger in the
     2015--16 season and also played for the
     University of Kentucky?
Answer: Karl-Anthony Towns

Context: Erin Wiedner (born in Mill Valley,
     California) is an American independent
     filmmaker, director, and
     cinematographer. Monte Hellman (born
     July 12, 1932) is an American film
     director, producer, writer, and editor.
Question: Erin Wiedner and Monte Hellman are
     both American film directors?
Answer: yes

Context: He rose to prominence with roles in
     the ensemble comedy-dramas "Dil Chahta
     Hai" (2001) and "Kal Ho Naa Ho" (2003).
     Kal Ho Naa Ho (English: "Tomorrow May
     Never Come"), abbreviated as KHNH, is a
     2003 Indian romantic drama directed by
     debutant director Nikkhil Advani.
Question: What is the abbreviation for the
     movie which starred Saif Ali Khan in
     2003?
Answer: KHNH
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